SESSION 1 · TUE 20 OCTOBER 2026

Portfolio Theory and Its Limits

Course overview, diversification and estimation error

Portfolio Management (28E34600), Autumn 2026
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Reading: lecture notes Ch. 1–5 (review); pre-reading package

Agenda

1. **The course: how lectures and cases fit together**
2. **Portfolio theory in numbers: what do μ, Σ and w mean?**
3. **60/40: why it works and when it fails**
4. **Beyond 60/40: can we diversify better?**
5. **The weak link: inputs are estimates**
6. **Wrap-up and next steps**

Part 1

1 **The course: how lectures and cases fit together**

2 Portfolio theory in numbers: what do μ, Σ and w mean?

3 60/40: why it works and when it fails

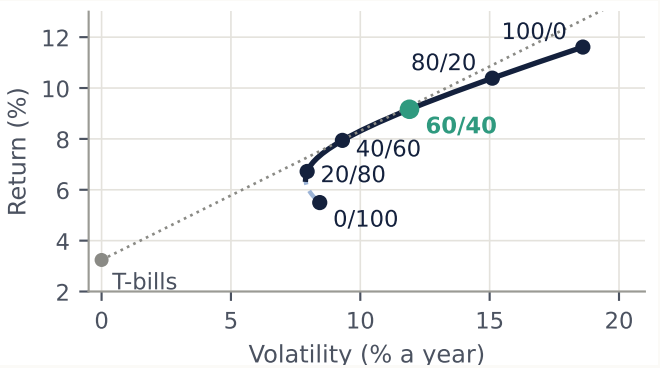
4 Beyond 60/40: can we diversify better?

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6 Wrap-up and next steps

The starting point: Markowitz's efficient frontier

Every stock–bond mix has an expected return and a risk



- **Frontier:** the highest expected return for each level of risk
- **Diversification** bends the curve to the left: less risk for the same return
- **With a risk-free asset**, the best risky mix is the one with the steepest line from T-bills: the highest Sharpe ratio

Markowitz (1952). Expected returns and risks estimated from US monthly data 1926–2025 (arithmetic averages, annualized); stocks: CRSP value-weighted; bonds: long-term government bonds.

What this course is about

The theory is simple; putting it to work is not.



Part 1: Asset allocation

A long-term investor chooses weights for asset classes. Case 1.

Part 2: Equity portfolios

A fund manager chooses weights for individual stocks. Case 2.

Part 3: The total portfolio

Risk parity, private assets, and the Finnish pension system.

Twelve sessions in three parts

Part 1: Asset allocation (Case 1)

1	Tue 20.10	Portfolio theory and its limits
2	Wed 21.10	Strategic asset allocation
3	Tue 27.10	Expected returns I (Ilmanen)
4	Wed 28.10	Expected returns II
5	Tue 03.11	Black–Litterman: SAA and TAA
6	Wed 04.11	Online: Case 1 clinic

Part 2: Equity portfolios (Case 2)

7	Tue 10.11	The equity problem; Case 2
8	Wed 11.11	From ideas to portfolios (Alasalmi)
9	Tue 17.11	Responsible investing
10	Wed 18.11	Online: Case 2 clinic

Part 3: The total portfolio and the Finnish pension system

11	Tue 24.11	Beyond 60/40; ATP
12	Wed 25.11	Finnish pensions; example exam

Lectures Tue and Wed 08:30–10:00. Clinics with Mikko Kauppila: online, times in MyCourses. Every lecture has slides and study notes; Session 3 has study notes for Case 1.

Two group cases: the applied work of the course

Case 1: Asset allocation — 15%

You advise the investment committee of a long-term euro investor with nine asset classes: expected returns, a risk model, a strategic allocation, tactical views with Black–Litterman, and a recommendation.

Report due Fri 13.11, 16:00; presentations in the week of 16.11.

Case 2: Equity fund — 25%

Your group launches a new fund (ESG, long-only or long/short) and pitches it to seed investors: signals, risk model, portfolio construction and an honest out-of-sample test.

Test plan Fri 20.11; **report and pitch Thu 26.11, 16:00**.

Groups of four. Register your group in MyCourses by **Wednesday 21 October, 16:00**.

Assessment and how to use the materials

Assessment

Case 1	15%
Case 2	25%
Examination	60%

Open-book exam on computers: course materials allowed, no other websites, no generative AI. Applied questions.

Materials (all in MyCourses)

- **Lecture notes:** the reference; read the chapters before each lecture
- **Slides:** written so that they can be read on their own
- **Case notes:** real managers, for Part 2
- **Pre-reading package:** what we assume you know, with exercises and a short coding warm-up
- **Polls in class:** scan the QR code; answers are anonymous

Poll 1: warm-up

A 60/40 portfolio puts 60% of the money in stocks. What share of its risk comes from stocks?

A about 40%

B about 60%

C about 75%

D about 90%



Scan to vote

Part 2

1 The course: how lectures and cases fit together

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3 60/40: why it works and when it fails

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The portfolio problem

Return and risk of a portfolio

$$\mu_p = \boldsymbol{w}^\top \boldsymbol{\mu}, \qquad \sigma_p^2 = \boldsymbol{w}^\top \boldsymbol{\Sigma}\, \boldsymbol{w}$$

Weights $\boldsymbol{w}$, expected returns $\boldsymbol{\mu}$, covariance matrix $\boldsymbol{\Sigma}$.

The investor's choice

$$\max_{\boldsymbol{w}} \; \boldsymbol{w}^\top \boldsymbol{\mu} - \frac{\gamma}{2}\, \boldsymbol{w}^\top \boldsymbol{\Sigma}\, \boldsymbol{w}$$

γ: how much the investor dislikes risk.

- Risk depends on **covariances**, not only on each asset's own risk
- The solution is $\boldsymbol{w}^* = \frac{1}{\gamma}\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$: weights are only as good as $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$
- The whole course is about where $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ come from, and what to do when they are wrong

The Greek letters in numbers: stocks and bonds

US stocks and long-term government bonds, 1926–2025, annualized

Inputs

	Stocks	Bonds
Expected return μ	11.6%	5.5%
Volatility σ	18.6%	8.4%

Correlation $\rho = 0.078$

Weights $\boldsymbol{w}$: 60/40

$$\boldsymbol{w} = \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix}, \qquad \boldsymbol{\mu} = \begin{bmatrix} 0.116 \\ 0.055 \end{bmatrix}$$

Covariance matrix Σ

$$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_S^2 & \rho\,\sigma_S\sigma_B \\ \rho\,\sigma_S\sigma_B & \sigma_B^2 \end{bmatrix} = \begin{bmatrix} 0.0346 & 0.0012 \\ 0.0012 & 0.0071 \end{bmatrix}$$

Diagonal: variances, $0.186^2 = 0.0346$ and $0.084^2 = 0.0071$.
Off-diagonal: covariance, $0.078 \times 0.186 \times 0.084 = 0.0012$.

Two assets need 2 expected returns, 2 volatilities and 1 correlation. Fifty assets need 50, 50 and 1,225.

Arithmetic averages and volatilities of monthly total returns, annualized. A three-asset example with stylized inputs is in lecture notes Ch. 4.

The 60/40 portfolio, step by step

Expected return $\mu_p = w^\top \mu$

$0.6 \times 11.6\% + 0.4 \times 5.5\% = \mathbf{9.2}\%$

Variance $\sigma_p^2 = w^\top \Sigma\, w$

Stocks	$0.6^2 \times 0.0346$	0.0125
Bonds	$0.4^2 \times 0.0071$	0.0011
Covariance	$2 \times 0.6 \times 0.4 \times 0.0012$	0.0006
Total		**0.0142**

$\sigma_p = \sqrt{0.0142} = \mathbf{11.9}\%$

11.9%

60/40 volatility

14.5%

weighted average of the two volatilities,
$0.6 \times 18.6\% + 0.4 \times 8.4\%$

The gap is diversification. But look at the variance: the stock term alone is 0.0125 of 0.0142. We come back to this in the answer to Poll 1.

Part 3

1 The course: how lectures and cases fit together

2 Portfolio theory in numbers: what do μ, Σ and w mean?

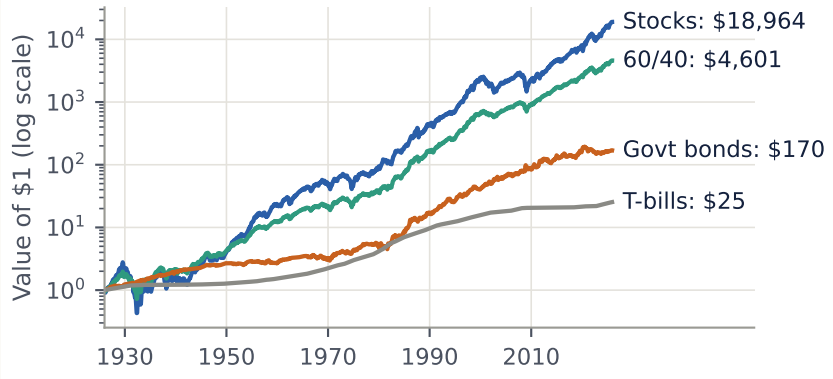
3 **60/40: why it works and when it fails**

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A century of returns: stocks earned the most, and swung the most



US data, January 1926 to December 2025, monthly, before costs and taxes. Stocks: CRSP value-weighted index; bonds: long-term US government bonds; 60/40 rebalanced monthly. Data: Goyal and Welch (2008), updated to 2025.

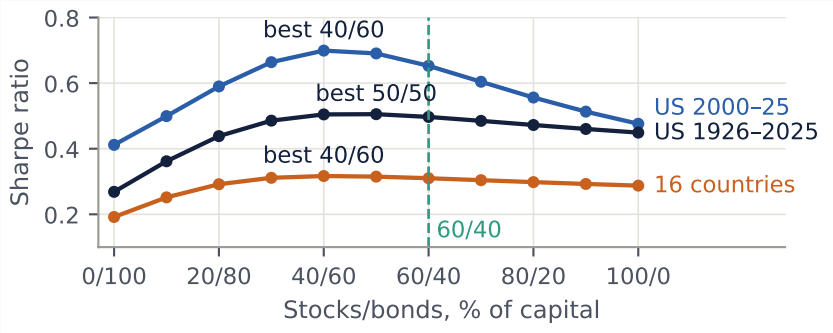
What one hundred years of data say

US, 1926–2025	Return a year	Real return	Volatility	Sharpe ratio
Stocks	10.4%	7.2%	18.6%	0.45
Long-term government bonds	5.3%	2.3%	8.4%	0.27
Treasury bills	3.3%	0.3%	0.9%	–
60/40 stocks and bonds	**8.8%**	**5.7%**	**11.9%**	**0.50**
16 countries, 1871–2020: was the US typical?				
Stocks	–	4.6%	22.0%	0.29
60/40 stocks and bonds	–	3.8%	15.3%	0.31

- US stocks paid about **8 points a year over bills**; 60/40 kept most of it with **two-thirds of the risk**
- The US was **exceptional**: across 16 countries, real stock returns were 4.6% a year, not 7.2%

Return a year: geometric average. Real: deflated by CPI. US: monthly excess returns over bills, annualized. 16 countries: JST Macrohistory database (Jordà et al. 2019), annual, pooled country-years; nominal returns not comparable across currencies.

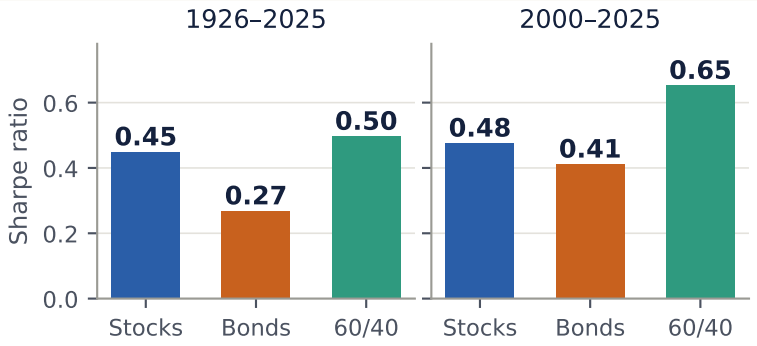
Which mix had the highest Sharpe ratio?



The curve is flat at the top: 40/60 to 60/40 are almost equally good. Country by country, the best mix ranges from 20/80 to 100/0.

Excess returns over bills; bonds are long-term government bonds. 16 countries: JST Macrohistory database (Jordà et al. 2019), annual, pooled.

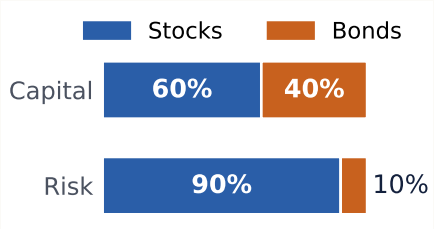
Why 60/40 became the standard: a better Sharpe ratio



Since 2000, bonds rose when stocks fell. That made 60/40 look even better: Sharpe ratio 0.65.

US monthly excess returns over Treasury bills; Sharpe ratio = average excess return / volatility, annualized.

Poll 1 revealed: 60% of the money, 90% of the risk



Stocks are more than twice as volatile as bonds and barely correlated with them, so each dollar in stocks carries far more risk.

Answer D.

Risk contributions: the 60/40 numbers again

1. Covariance of each asset with the portfolio, $\boldsymbol{\Sigma w}$:

Stocks: $0.6 \times 0.0346 + 0.4 \times 0.0012 = 0.0212$
Bonds: $0.6 \times 0.0012 + 0.4 \times 0.0071 = 0.0036$

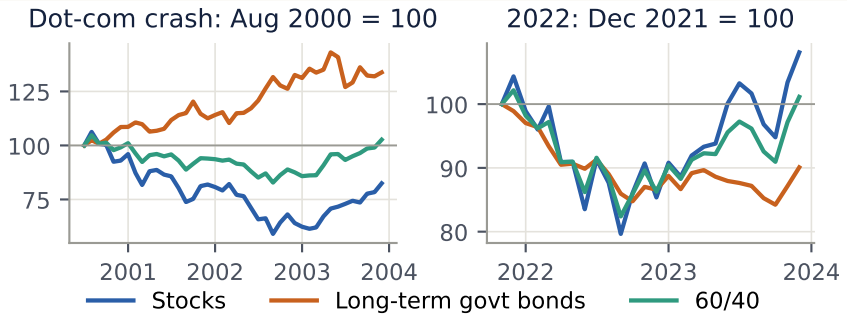
2. Risk contribution $RC_i = w_i\,(\boldsymbol{\Sigma w})_i / \sigma_p$:

Stocks: $0.6 \times 0.0212/0.119 = 10.7\%$
Bonds: $0.4 \times 0.0036/0.119 = \ \ 1.2\%$

3. They add up to $\sigma_p = 11.9\%$. Shares of risk: $10.7/11.9 =$ **90%** and $1.2/11.9 =$ **10%**.

Inputs from the "Greek letters" slide: US stocks and long-term government bonds, monthly returns 1926–2025, annualized.

Two crashes: 60/40 was an equity bet



Aug 2000 – Sep 2002: stocks −44%, bonds +29%, 60/40 −21%. Bonds did their job, yet stock risk dominated.

Dec 2021 – Sep 2022: stocks −24%, bonds −13%, 60/40 −19%. Inflation and rising rates hit both.

Value of each portfolio, monthly. Left: August 2000 = 100. Right: December 2021 = 100. US data as before.

The stock–bond correlation is not a constant



Average correlation of excess returns: 1970–1999 +**0.35**; 2000–2021 −**0.28**; 2022–2025 +**0.57**.

36-month rolling correlation of monthly excess returns, US stocks and long-term government bonds.

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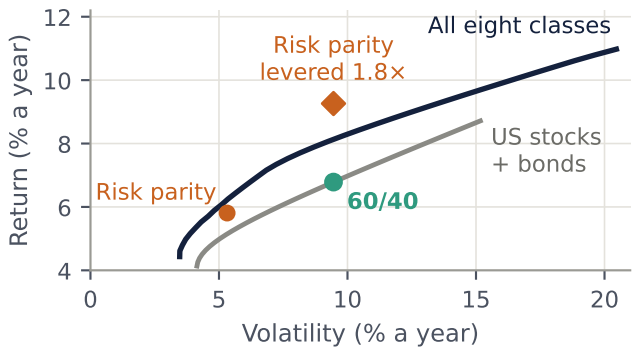
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New practices: putting theory to work better

Can we push the frontier up by diversifying beyond stocks and bonds?



- Case 1 data, 2000–2025, the eight risky classes: best Sharpe ratio 0.98, up from 0.73
- Risk parity at 60/40 risk: Sharpe 0.83 vs 0.57
- **But:** adding assets *always* lifts a frontier fitted to history

Case 1 data, 2000–2025, USD; long-only frontiers from historical means and covariances (in sample). Risk parity: equal risk contributions, financed at the cash rate.

Three new practices: each solves one problem and creates another

	Endowment model	Risk parity	Total portfolio approach
Examples	Yale; Varma	All Weather (1996)	NZ Super, CPP, GIC
Idea	private assets, hedge funds; few bonds	equal risk; lever up low-risk assets	one reference portfolio; assets compete for capital
Needs	patience, top managers	leverage, stable correlations	strong governance, skilled staff
Weak spot	illiquidity; smoothed prices hide risk	2022: stocks and bonds fell together	unclear accountability

All three still need the same inputs: expected returns and risk.

Swensen (2000); Qian (2005); Asness, Frazzini and Pedersen (2012); AQR (2026).

What the track records say

	Return a year	What else to know
Endowment: **Yale**, FY2009–25, USD	8.3%	−24.6% in FY2009, +40.2% in FY2021
Endowment: **Varma vs Keva**, 2009–25, EUR	6.5% vs 7.1%	volatility 6.0% vs 6.7%; 2022: −4.9% vs −7.0%
Risk parity: **our backtest**, 2000–25	9.2% vs 6.5% for 60/40	in sample, before costs; 2022: −7% vs −16%
TPA: **NZ Super Fund**, since 2003	10.1% vs 8.7% (80/20)	+1.4 points a year over its reference
TPA: **CPP Investments**, 10 years	8.3%	+1.4 points over benchmarks; −1.6 in FY2025

Lower measured risk and some value added, but smoothed prices and short records.

Fiscal years: Yale to June, NZ Super to June 2025, CPP to March 2025. Varma, Keva: Tela, calendar years. Risk parity: Case 1 data.

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Historical returns are not expected returns

Forward-looking expected real returns are well below the past century

Real return a year	US history 1926–2025	AQR forecast, end-2025
US stocks	7.2%	3.9%
US government bonds	2.3%	2.4% (10-year)
US cash	0.3%	1.3%
60/40	5.7%	3.4% (global)

- High past stock returns came partly from **rising valuations**, which cannot repeat forever
- Forecasts start from today's yields and valuations: Sessions 3 and 4

AQR (2026), *Capital Market Assumptions for Major Asset Classes*, Exhibit 1: expected real returns, local currency, as of December 2025. History: geometric averages, CPI-deflated.

Poll 2

Over 100 years, stocks beat bills by 8.4% a year on average. How precise is this estimate (one standard error)?

A ±0.2 percentage points

B ±0.5 percentage points

C ±1 percentage point

D ±2 percentage points



Scan to vote

Poll 2 revealed: even a century of data is short

Standard error of an average

$$\mathrm{se}(\hat{\mu}) = \frac{\sigma}{\sqrt{T}} = \frac{18.6\%}{\sqrt{100}} \approx 1.9\%$$

Answer D. The 95% interval runs from about 4.7% to 12.0%.

- With 25 years of data: ±3.7 points
- More frequent data do not help: the average depends on the length of the sample, not the number of observations
- Volatility is estimated far more precisely than the mean

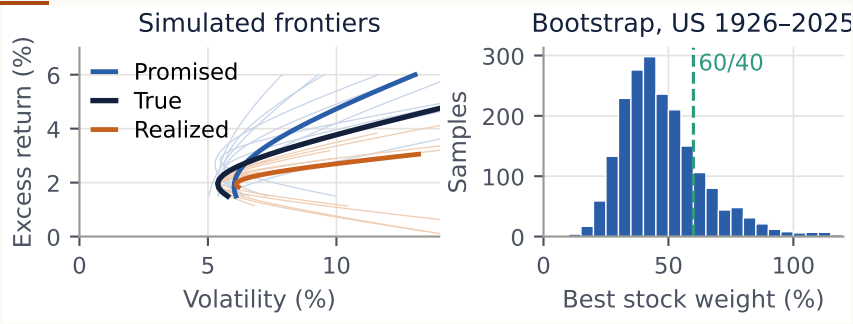
Expected returns are the inputs we know least well, and the optimizer is most sensitive to them.

The optimizer takes your errors literally



Simulation: 1,000 samples of 30 years of monthly returns from known true parameters; dots are the true optimal weights $\boldsymbol{w}^* = \frac{1}{\gamma}\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$ ($\gamma = 4$), bars the 5th–95th percentiles of the weights estimated from each sample. All 20 stocks have the same true expected return.

The frontier itself is estimated: weights are random variables



Left: the optimizer promises 4% at 8.4% risk; the true risk is 10.9%, and the portfolios earn 2.8%.

Right: resampling a century, the best stock weight is 26–81%.

Broadie (1993); Michaud (1998). Left: 200 simulated 30-year samples (thin lines: 12 examples; thick: averages). Right: 2,000 bootstrap samples of calendar years, US excess returns 1926–2025.

Why correlated securities maximize the error

Broad asset classes

Few assets, different risks, low correlations. Errors move the weights, but the portfolio stays recognizable.

Individual stocks

Many assets, high correlations. $\boldsymbol{\Sigma}^{-1}$ treats near-identical stocks as a "free" long–short trade and bets on tiny, noisy differences in $\hat{\boldsymbol{\mu}}$.

- Michaud (1989): mean–variance optimizers are "estimation-error maximizers"
- DeMiguel, Garlappi and Uppal (2009): the simple $1/N$ portfolio is hard to beat out of sample
- In Part 2 we never feed raw return estimates for stocks to an optimizer

What practitioners do about it, and where we do it in this course

Remedy	Idea	Session
Forward-looking expected returns	yields, growth and valuation instead of history	3, 4
Reference portfolio and limits	start from a sensible portfolio; constrain deviations	2, 5
Shrinkage and robustness checks	pull noisy inputs toward simple targets	2, 5
Black–Litterman	equilibrium returns plus views, weighted by confidence	5
Factor risk models	structure $\boldsymbol{\Sigma}$ for many stocks	7
Calibrated alphas, costs	scale forecasts down; trade only when it pays	7, 8
Out-of-sample tests vs $1/N$	judge the whole process on unseen data	8, Case 2

Part 6

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Takeaways

Diversification is about risk

60/40 is 60% stocks by capital and about 90% by risk; it fell 21% in the dot-com crash.

Correlations change

Bonds hedged stocks from 2000 to 2021, but not in 1970–1999 or in 2022.

Expected returns are estimates

A century of data pins the equity premium down only to about ±2 points.

Optimizers maximize errors

Fine with a few broad asset classes, dangerous with many correlated stocks. Treat weights as estimates too.

Before Wednesday

- **Register your group** of four in MyCourses by **Wed 21.10, 16:00**
- **Read** lecture notes Ch. 6–9 (Session 2: the Goldman Sachs case and strategic allocation); review Ch. 1–5
- **Work through** the pre-reading exercises and the coding warm-up
- **Look at** the Case 1 brief and data: Case 1 starts now

Next: Session 2, Wed 21.10 – Strategic asset allocation: the Goldman Sachs case and the nine-asset problem.

References (backup)

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